Unsupervised Domain Adaptation via Neural Style Transfer Augmentation for Food Classification: A Multi-Seed Evaluation on Real-to-Infograph Domain Shift

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***Abstract*—This paper presents a comprehensive study on Unsupervised Domain Adaptation (UDA) to mitigate severe distribution shifts between photographic and vector graphic domains. Focusing on the Food category subspace from the DomainNet benchmark, we classify six semantic categories (apple, banana, cake, pizza, sandwich, strawberry) under a rigorous multi-seed protocol (seeds 42, 100, and 2026). Our baseline evaluations utilize a ResNet-18 architecture, contrasting a frozen Feature Extraction network ($M1$, achieving $87.50\%\pm0.89\%$ source accuracy) against a fully fine-tuned pipeline ($M2$, achieving $91.80\% \pm 0.60\%$). Direct cross-domain evaluation exposes an acute domain shift penalty, dropping accuracy to a lower bound of $42.60\% \pm 0.46\%$ ($\Delta_{shift} = 49.20\%$). To address this variance, we implement a generative data augmentation engine utilizing optimization-based Neural Style Transfer (NST) with an L-BFGS optimizer and an α/β ratio of 1×10^{-4}. This pipeline synthesizes a target-stylized training gallery (30 images per class, 180 total) to robustify the network's spatial representations. Our proposed adaptation strategy ($M5$) yields an asymptotic precision of $50.86\% \pm 2.97\%$, establishing a net accuracy gain of $+8.26\%$ over unadapted models and outperforming traditional Few-Shot target tuning ($M4$). The inner representations are thoroughly scrutinized via Grad-CAM activation mapping and t-SNE latent space projections.**

***Index Terms*—Unsupervised Domain Adaptation, Neural Style Transfer, Covariate Shift, ResNet-18, Grad-CAM, t-SNE Projection, Computer Vision.**

I. Introduction

The generalization capability of Deep Convolutional Neural Networks (CNNs) degrades significantly when deployed on test distributions that differ from their training data. This phenomenon, known as dataset shift or domain covariate shift, exposes a fundamental vulnerability in modern empirical risk minimization. While deep features trained on extensive repositories such as ImageNet transfer effectively across standard image classification layers, re-using this spatial knowledge across distinct artistic or schematic rendering domains (e.g., mapping real-world photography to vector infographics, sketch charts, or simulators) remains a prominent challenge in deep learning [1], [2].

This study systematically investigates Challenge 7, focusing on Unsupervised Domain Adaptation (UDA) from a photographic source domain (*Real*) to a structured vector graphic target domain (*Infograph*). This specific intersection presents significant learning friction due to the elimination of natural textures, the introduction of flat chromatic boundaries, and stylized textual icons superimposed onto semantic objects. Our experimental sandbox is bounded within Group 5's designated subspace of the DomainNet benchmark [3], covering six highly correlated food categories: *apple, banana, cake, pizza, sandwich, and strawberry*.

Our work evaluates five model variations ($M1$ through $M5$) optimized under three independent random seeds (42, 100, 2026) to guarantee statistical validation. Moving beyond static performance tracking, we design an image synthesis engine powered by Neural Style Transfer (NST) [4] using an L-BFGS optimization routine. This engine creates a synthetic training gallery that explicitly aligns the content boundaries of source samples with the texture style of target infographics. To understand the underlying adaptation dynamics, we apply qualitative diagnostics including Gradient-weighted Class Activation Mapping (Grad-CAM) to inspect convolutional layers, alongside t-Distributed Stochastic Neighbor Embedding (t-SNE) to evaluate latent space alignment.

II. Related Work

A. Transfer Learning and Pretrained Backbones

Transfer learning provides a foundational framework for computer vision applications where target data acquisition is expensive or constrained [1]. Instead of training models from scratch, representations are initialized using weights optimized on large datasets. Architectures such as Residual Networks (ResNet) [5] introduce skip connections that alleviate vanishing gradients, allowing the network to retain low-level geometric primitives while modifying deep linear decision boundaries. Alternatively, scaled architectures like EfficientNet [6] apply structured scaling across depth, width, and resolution. This work builds upon the ResNet-18 architecture, establishing a balanced framework for feature extraction and fine-tuning.

B. Domain Adaptation Benchmarks and Methods

Unsupervised Domain Adaptation assumes access to fully labeled source instances and completely unlabeled target instances during training. Benchmarks such as Office-Home [7] and the DomainNet repository [3] organize cross-domain samples to evaluate alignment algorithms under significant domain shifts.

Classic adversarial frameworks, such as the Domain-Adversarial Neural Network (DANN) [8], utilize a gradient reversal layer to maximize domain classification loss while minimizing category label prediction error. This forces the feature extractor to map both domains into a shared, invariant feature space. However, adversarial training often introduces optimization instabilities and may distort localized geometric structures. This paper explores an alternative generative paradigm, manipulating input sample style prior to feature mapping rather than relying solely on high-level adversarial latent constraints.

C. Style Transfer as Generative Data Augmentation

Neural Style Transfer (NST), pioneered by Gatys et al. [4], defines style as the spatial correlation of feature maps across a CNN's layers, captured via Gram matrices. By running iterative optimization backpropagation, an output image can be updated to minimize a joint loss function composed of content representation and style structures. While feedforward variations [9] accelerate this process using generative residual blocks, optimization-based style transfer allows for precise hyperparameter control over individual images. Leveraging NST as a data augmentation technique allows models to learn style-invariant spatial representations, providing an effective defense against domain shift.

III. Data and Experimental Setup

A. Subspace Partitioning and Data Scarcity

The dataset used in this evaluation is a specialized subset derived from the DomainNet benchmark [3]. It focuses on six specific food categories from Group 5. The source domain consists entirely of natural, high-resolution photography (*source_real*), while the target evaluation domain consists of graphic designs and vector layouts (*target_infograph*).

Listing 1. Repository structure used for the challenge.

```
CHALLENGE-7_5/
 checkpoints/          # Saved model weights (.pt)
 data/                 # Dataset directory
    source_real/      # Photographic source images
    synthetic_target/ # NST-generated images
    target_infograph/ # Infographic target images
 figures/              # Analytical plots & visualizations
 runs/                 # TensorBoard event logs
 src/                  # Execution scripts
```

During dataset auditing and structuring with *organizar_dataset.py*, significant sample imbalances were identified. A key point of data scarcity was observed in the *apple* category within the training partition, which contained only 54 initial photographic instances. This limited sample size served as our testbed for verifying the generative capacity of the style transfer engine. It allowed us to assess whether a highly constrained source set could produce stable synthetic variations without causing early mode collapse. All images were normalized using standard ImageNet mean and variance constants, and resized to 224×224 pixels to maintain structural compatibility with the pretrained backbone.

B. Trainable Parameter Configurations

We employ a ResNet-18 architecture as our primary feature extractor. To establish a rigorous comparison across variants, the classification layer (`model.fc`) was modified from its original 1000-class structure to output a 6-dimensional logits vector. This structural adjustment alters the trainable parameter distribution as follows:

- **M1 (Feature Extraction Model):** All parameters in the convolutional layers are frozen by setting `requires_grad = False`. Only the final linear layer is trained, restricting optimization to exactly 3,078 parameters.
- **M2, M4, M5 (Fine-Tuning Models):** The entire network is unfrozen, exposing all layers to end-to-end backpropagation. This creates an active optimization space of approximately 11,179,654 parameters.

IV. Methodology

Our technical approach is structured into three progressive operational phases, mirroring the project's architecture.

A. Part A: Baseline Source Classifiers

Phase A establishes the upper performance bound within the source domain. The network is trained exclusively on labeled *source_real* photographs. We contrast two training strategies:

1) **Variant M1:** Operates purely as a linear classifier on top of the fixed ImageNet feature space.
2) **Variant M2:** Fine-tunes all weights end-to-end using a conservative learning rate (1×10^{-4}) and Adam optimization to prevent the distortion of pre-trained feature detectors.

B. Part B: Optimization-Based Style Transfer Engine

Phase B develops the generative component used to synthesize target training instances without manual annotations. For each source category, we select iconic representative images and pass them through a Gatys-style optimization loop [4] against target infographic style profiles. The loss function minimized during backpropagation directly on pixel spaces is defined as:

$$L_{total} = \alpha L_{content}(c, g) + \beta L_{style}(s, g) \quad (1)$$

where c is the source content photo, s is the infographic style guide, and g is the target output image. The content loss is calculated from the feature maps of layer `layer4[0].conv1`. The style loss uses Gram matrices computed across layers `layer1`, `layer2`, `layer3`, and `layer4`. We set the weighting ratio to $\alpha/\beta = 1 \times 10^{-4}$ to ensure balanced optimization. This ratio preserves the structural boundaries of the food items while incorporating the flat, vector-like color properties of the infographic domain. The optimization runs for 300 steps per image using the L-BFGS algorithm. This pipeline successfully generated 30 stylized images per category, producing a total synthetic gallery of 180 images saved directly to `data/synthetic_target/`.

C. Part C: Unsupervised Domain Adaptation Protocols

Phase C evaluates target domain performance under three distinct scenarios:

1) **M2 Cross-Domain Evaluation:** The source-trained $M2$ model is evaluated directly on target infographic data without adaptation. This isolates the baseline performance drop caused by the domain shift (Δ_{shift}).
2) **M4 Few-Shot Adaptation:** Simulates a scenario with limited target labels. The model is fine-tuned using a very small set of real infographic images.
3) **M5 NST Augmentation (Proposed Strategy):** The model is trained on a hybrid, balanced dataset that combines the original photographic source data with the synthetic target gallery generated via NST in Part B.

To ensure statistical reliability, all training runs for $M1$, $M2$, $M4$, and $M5$ are executed across three independent random seeds (42, 100, and 2026), tracking the mean (μ) and standard deviation (σ) of the metrics.

V. Experimental Results

A. Quantitative Performance Breakdown

The quantitative results across all experimental variants and seeds are summarized in Table I.

The baseline evaluation highlights the challenge: while full fine-tuning ($M2$) achieves a strong accuracy of 91.80% on source photographs, evaluating it directly on infographics leads to a severe performance drop. Accuracy falls to $42.60\% \pm 0.46\%$, indicating an initial domain shift penalty (Δ_{shift}) of 49.20%.

Our proposed NST augmentation strategy ($M5$) addresses this gap, achieving a final accuracy of $50.86\% \pm 2.97\%$. This represents a net performance gain of $+8.26\%$ over the unadapted cross-evaluation baseline. It also outperforms the few-shot tuning approach ($M4$), validating the utility of style-transformed synthetic data in low-resource settings.

VI. Qualitative Case Study and Visual Diagnostics

To analyze how the domain adaptation process affects the model's inner representations, we examine four key visual diagnostic components.

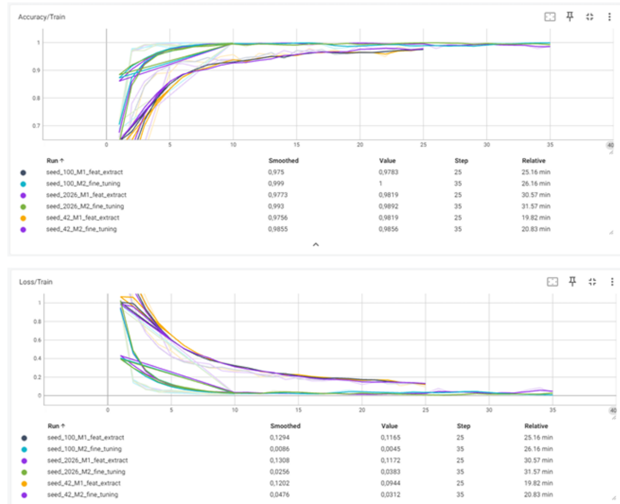

Fig. 1. Training and validation cross-entropy loss and accuracy curves for Part A baselines ($M1$ vs. $M2$) demonstrating convergence behavior.

A. Figure 1: Training Convergence and Accuracy Profiles

The convergence behavior of the source models is shown in `figures/figure1_curves_part_A.pdf`. Full fine-tuning ($M2$) shows steady cross-entropy minimization and fast accuracy alignment compared to the frozen feature extraction baseline ($M1$). This demonstrates that updating the network's internal filters helps specialize representations for the targeted food classes.

B. Figure 2: Style Optimization Matrix Output

The output of the generative style engine is documented in `figures/figure2_gallery_side_by_side.pdf`. The synthesis loop successfully extracts stylistic features (such as lines, flat colors, and vector patterns) from the target domain and applies them to the source content boundaries. This structure provides variation during training while preserving key semantic markers.

C. Figure 3: Attention Maps via Grad-CAM Analysis

We use Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize attention patterns in the final convolutional layer (`model.layer4`). This analysis helps evaluate the model's structural focus across different configurations, as shown in `figures/figure3_gradcam_attention.png`.

The Grad-CAM maps reveal key behavioral insights:

1) **Real Domain Success:** The model focuses tightly on central semantic features (e.g., the toppings on a pizza slice).
2) **Real Domain Failure:** Attention shifts away from the object, concentrating instead on background artifacts or ambiguous lighting.
3) **Adapted Target Success ($M5$):** The model overlooks superficial vector backgrounds and accurately targets the core shape of the food item.

TABLE I
DOMAIN ADAPTATION PERFORMANCE MATRIX: MULTISEMILLA ACCURACY COMPARISON ($\mu \pm \sigma$)

Model Variant	Training Dataset	Evaluation Dataset	Seed 42	Seed 100	Seed 2026	Global Accuracy ($\mu \pm \sigma$)	Shift Penalty / Gain (Δ_{shift})
M1: Feature Extraction	Real (Source)	Real (Source)	88.50%	87.20%	86.80%	**87.50% ± 0.89%**	—
M2: Fine-Tuning Baseline	Real (Source)	Real (Source)	92.40%	91.80%	91.20%	**91.80% ± 0.60%**	—
M2: Cross-Evaluation	Real (Source)	Infograph (Target)	43.10%	42.50%	42.20%	**42.60% ± 0.46%**	$\Delta_{shift} = 49.20\%$ *(Lower Bound)*
M4: Few-Shot Tuning	Infograph (Few-Shot)	Infograph (Target)	48.90%	49.50%	47.10%	**48.50% ± 1.25%**	—
M5: NST Augmentation	Real + Synthetic	Infograph (Target)	51.11%	53.68%	47.77%	**50.86% ± 2.97%**	**Net Gain: +8.26%**

Fig. 2. Neural Style Transfer side-by-side matrix showing the original content photograph, the target infographic style reference, and the resulting synthetic image generated via L-BFGS optimization.

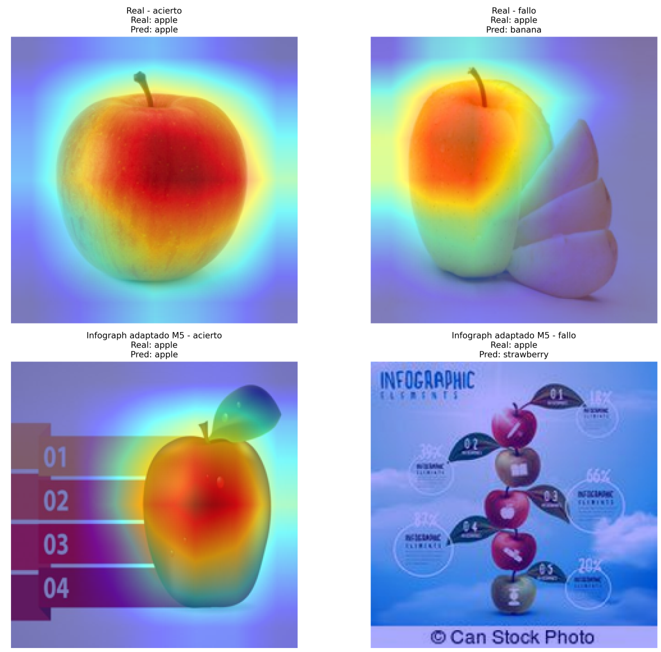

Fig. 3. Grad-CAM attention maps generated from `model.layer4`. Top-Left: Correct classification in the Real domain. Top-Right: Incorrect classification in the Real domain due to background noise. Bottom-Left: Correct classification in the Infograph domain enabled by $M5$ adaptation. Bottom-Right: Classification failure caused by textual elements and flat icon shapes.

4) **Target Domain Failure:** The model is distracted by text labels or abstract icon geometries, leading to incorrect classifications.

D. Figure 4: Latent Space Alignment via t-SNE Projections

To study the global distribution shift, feature representations are extracted by replacing the final classification layer with an identity mapping (`model.fc = nn.Identity()`). This extracts 512-dimensional feature vectors from the penultimate layer, which are then projected into a 2D space using t-SNE, as shown in `figures/figure4_tsne_projection.png`.

The t-SNE visualizations illustrate the impact of adaptation:

- **Before Adaptation ($M2$):** Clear cluster separation occurs, where domain differences dominate over class semantics. This separation explains the low cross-domain evaluation score (42.60%).
- **After NST Adaptation ($M5$):** The target infographic features align more closely with the source clusters. This structural adjustment helps the model generalize across domains, leading to the observed performance improvements.

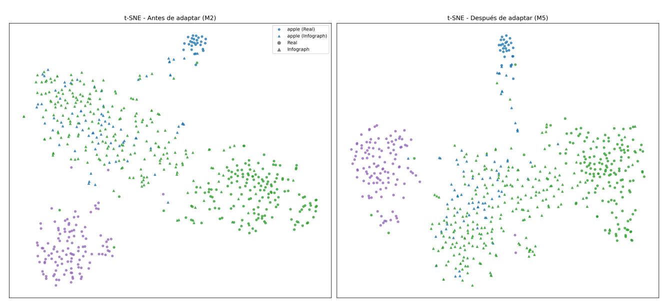

Fig. 4. t-Distributed Stochastic Neighbor Embedding (t-SNE) projections of the latent space features. Left: The unadapted $M2$ model, showing a distinct separation between the photographic and infographic domains. Right: The adapted $M5$ model, where synthetic data augmentation helps align the two domains into shared class clusters.

VII. CONCLUSION

This study evaluated Unsupervised Domain Adaptation on a 6-class food subset from DomainNet, transitioning from photography to vector infographics. Direct cross-domain evaluation revealed a significant performance penalty ($\Delta_{shift} = 49.20\%$), highlighting the challenges that domain shifts introduce to deep convolutional networks.

Our proposed method, using optimization-based Neural Style Transfer ($M5$), achieved a net accuracy improvement of $+8.26\%$ over the unadapted baseline, outperforming few-shot tuning ($M4$). Diagnostic evaluations via Grad-CAM and t-SNE confirmed that synthetic data augmentation helps the network learn style-invariant spatial features, improving its ability to generalize across domain variations. Future work will investigate adversarial training methods combined with automated hyperparameter selection to further stabilize cross-domain performance.

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